

# Pol del Castillo

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## Summary

M.S. Mechanical Engineering candidate at RPI researching robot learning, with hands-on experience across controls, embedded firmware, and sensing. Seeking robotics and controls engineering roles, with an interest in medical devices.

## Education

**Rensselaer Polytechnic Institute** | *M.S. Mechanical Engineering, GPA 3.77* Troy, NY | Aug 2025 – May 2027  
**Georgia Southern University** | *B.S. Mechanical Engineering, GPA 3.79, Magna Cum Laude* Statesboro, GA | Aug 2021 – May 2025

## Experience

**ViaDose (early-stage startup, stealth)** | *R&D Hardware Engineer* Troy, NY | 2026 – Present

- Own the sensing architecture, microcontroller firmware, and energy budget for a connected medical-adherence device.
- Build the load-cell sensing chain and dose-detection algorithm: strain-gauge bridge, EMA and median filtering, CUSUM change detection.
- Bring up firmware on custom PCBs, integrating load cells and IMUs over I2C with BLE, NFC, and LTE-M on a coin-cell budget.

**Rensselaer Polytechnic Institute** | *Graduate Research Assistant, Wen Research Group* Troy, NY | Aug 2025 – Present

- Fine-tune NVIDIA GR00T and benchmark open VLA baselines (SmolVLA, Pi-0) via Hugging Face LeRobot on a 7-DOF Kinova arm.
- Build and calibrate a 5-DOF SO-101 arm, collect teleoperated demonstrations, and train ACT and diffusion-policy models.
- Implement a flow-matching (CondOT) policy from wrist-camera images and bridge Isaac Sim to ROS 2 Jazzy over Fast DDS for sim-to-real.

**Georgia Southern University** | *Undergraduate Research Assistant, Fire and Energy Lab* Statesboro, GA | 2024 – 2025

- Implemented a finite-volume immersed boundary method in Julia computing drag coefficients for vegetation at Re 140–307.
- Matched five published studies (1.29 against a 1.25–1.34 range); presented at the 2024 Student Research Symposium.
- Built a hybrid physics and neural-network model in Julia's SciML ecosystem learning lithium-ion degradation from data.

## Projects

**Automated Linear Scale System, Howmet Aerospace** | *Project Manager, Senior Capstone* Statesboro, GA | 2025

- Led an 11-person team across three subgroups to automate billet measurement on a production saw line in 16 weeks.
- Achieved 0.0008 in measurement precision, replacing a handheld tape measure accurate to roughly 1/16 in.
- Selected a ball-screw architecture via QFD (roughly \$1,000 against \$10,000) and built the Arduino, stepper, and HMI control stack.

**Acrobot Swing-Up and Balance** | *Adaptive Control and Reinforcement Learning* RPI | 2026

- Built a certainty-equivalence adaptive LQR in MATLAB: four parallel RLS estimators identify the model online each step.
- Paired the stabilizer with a DQN swing-up policy that hands off to LQR once the arm reaches the upright region.
- Stabilized the full nonlinear dynamics under a 10 N-m torque limit; mapped the basin of attraction over 400 simulations.

**DRAWbot, Vision-Guided Manipulation** | *Robotics Course, Team of 4* RPI | 2025

- Programmed a 5-DOF arm to reproduce sketches on paper, converting images to contours with OpenCV.
- Solved inverse kinematics per contour point with a Jacobian-inverse method on a product-of-exponentials model.
- Designed and 3D-printed a spring-loaded end-effector that keeps the pen in contact through surface variation.

## Technical Skills

**Robot learning:** PyTorch, LeRobot, GR00T, imitation learning, diffusion policies, flow matching, ACT, Isaac Sim / Isaac Lab, ROS 2  
**Embedded and hardware:** C/C++, Nordic nRF, Zephyr, Arduino, PlatformIO, I2C, SPI, UART, BLE, load cells, IMUs, PCB bring-up  
**Controls:** LQR, adaptive control, RLS system identification, inverse kinematics, Monte Carlo validation  
**Programming:** Python, Julia, MATLAB, SciML, Git **Languages:** English, Spanish

## Leadership and Athletics

**Georgia Southern Men's Tennis** | *NCAA Division I Student-Athlete, Team Captain* Statesboro, GA | 2021 – 2025

- Led the team as captain to the 2024 Sun Belt Regular Season Championship and its first year-end national ranking (No. 66 of 264).
- Named Georgia Southern Student-Athlete of the Year for 2023-24, the university's top athletic honor.
- Balanced 20+ hours per week of training and competition with a full-time engineering course load for four years.